

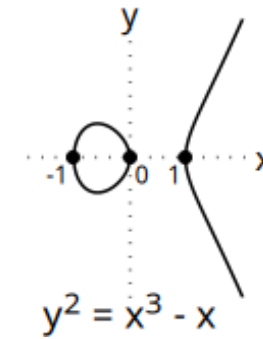
Fermat's last theorem for n=4
(any field K has been replaced with $\mathbb{Q}$)

Def * Elliptic curve over $\mathbb{Q}$ is equation of form:
 $y^2 = ax^3 + bx^2 + cx + d$ ($a, b, c, d \in \mathbb{Q}$, $a \neq 0$)
where RHS polynomial has no multiple root

Def For $a = \frac{m}{n}$ with $m, n \in \mathbb{N}$, $(m, n) = 1$
define height $H(a) := \max(|m|, |n|)$

(x, y) on elliptic curve
 $= (\frac{-1}{x}, \frac{y}{x^2})$ on as well,
 $H(x) = H(-1/x)$

Lem 1.5 For distinct $a, b, c \in \mathbb{Q}$ define $B, \tilde{C}$ and C
 $B = \{(u, v, w) \in \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q} \mid u^2 + a = v^2 + b = w^2 + c\}$
 $\tilde{C} = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid y^2 = (x-a)(x-b)(x-c)\}$.
 $C = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid y^2 = (x-a)(x-b)(x-c), y \neq 0\}$.
(1) $\exists$ mutually inverse maps $f: B \rightarrow C$, $g: C \rightarrow B$
 $f(u, v, w) = (u^2 + a + uv + vw + wu, (u+v)(v+w)(w+u),$
 $g(x, y) = (\frac{1}{2y}((x-a) - (b-a)(c-a)),$
 $\frac{1}{2y}((x-b) - (a-b)(c-b)), \frac{1}{2y}((x-c) - (a-c)(b-c)))$
(2) $\exists$ map $h: B \rightarrow \tilde{C}$, $h(u, v, w) = (u^2 + a, uvw)$



$$i: (x, y, z) \mapsto (\frac{y-x}{2}, \frac{z}{2}, \frac{x+y}{2})$$

$$j: (u, v, w) \mapsto (w-u, w+u, 2v)$$

$$\begin{cases} i: A_d \rightarrow B_d \\ j: B_d \rightarrow A_d \end{cases}$$

$$a=d, b=0, c=-d, \begin{cases} f: B_d \rightarrow C_d \\ g: C_d \rightarrow B_d \end{cases}$$

Lem 1.4 For $d \in \mathbb{Q}$ define:
 $A_d = \{(x, y, z) \in \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q} \mid x^2 + y^2 = z^2, \frac{1}{2}xy = d\}$
 $B_d = \{(u, v, w) \in \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q} \mid u^2 + d = v^2, v^2 + d = w^2\}$
 $C_d = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid y^2 = x^3 - d^2x, y \neq 0\}$.
 $\exists$ bijections between any 2 of A_d, B_d, C_d

$$\begin{cases} (i) & \Rightarrow A_d \neq \emptyset \\ (ii) & \Rightarrow B_d \neq \emptyset \\ (iii) & \Rightarrow C_d \neq \emptyset \end{cases}$$

Rem 1.6 $h \circ g: C \rightarrow \tilde{C}$ is called multiplication-by-2 map
for elliptic curve $y^2 = (x-a)(x-b)(x-c)$; image of $h \circ g$ is
 $\{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid y^2 = (x-a)(x-b)(x-c), x-a, x-b, x-c \text{ are } \square \text{ in } \mathbb{Q}\}$
because:
 $(x', y') = h(u, v, w) \Rightarrow x' = u^2 + a \Rightarrow x' - a \text{ is } \square \text{ in } \mathbb{Q}$
from definition of B (image of g): $u^2 + a = v^2 + b = w^2 + c$
 $x' - b = (v^2 + b) - b = v^2 \text{ is } \square \text{ in } \mathbb{Q}$
 $x' - c = (w^2 + c) - c = w^2 \text{ is } \square \text{ in } \mathbb{Q}$

$$a=1, b=0, c=-1$$

(ii) $x_0 - 1, x_0$ and $x_0 + 1$ are all squares
 $= \exists (x_1, y_1) \in C: h \circ g(x_1, y_1) = (x_0, y_0)$

(iii) $H(x_1) < H(x_0)$, $\nless$

Prop 0.13 The area of a right $\triangle$
whose sides are integers is not a $\square$

$$\neg(i) \quad d=1$$

Lem 1.3 $d \in \mathbb{Q}$, these are equivalent:
(i) $\exists$ rational number sides **right** $\triangle$ of area d
(ii) $\exists$ 3 $\square$ s of rational numbers that form
arithmetic progression of difference d
(iii) $\exists$ rational solution to $y^2 = x^3 - d^2x$
other than $(x, y) = (0, 0)$ and $(\pm d, 0)$

$$\neg(iii) \quad d=1$$

Prop 1.2 The only rational solutions
to $y^2 = x^3 - x$ are $(x, y) = (0, 0)$ and $(\pm 1, 0)$

$$1.1 \text{ sol} \cdot \frac{z^2}{y^6}, \nless$$

$$0 \neq \left(\frac{x^2 z}{y^3}\right)^2 = \left(\frac{z^2}{y^2}\right)^3 - \frac{z^2}{y^2}$$

Prop 1.1 $\nexists$ solution (x, y, z) to
 $x^4 + y^4 = z^4$ satisfying $xyz \neq 0$